Printed Pages: 02 **Sub Code: RCE-303**

Paper Id: 233408 **Roll No.** | | | | | | | | | | |

B.TECH
(SEM III) THEORY EXAMINATION 2022-23
FLUID MECHANICS

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

(a) Explain buoyancy and center of pressure.
(b) Define velocity potential function.
(c) What is the difference between notches and weir?
(d) List various types of minor losses.
(e) Explain boundary layer thickness with diagram.
(f) What are distorted models?
(g) Define dynamic similarity.

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

(a) Derive an equation to calculate total force acting on inclined plane.
(b) Define stream function & velocity potential function. Calculate for the velocity at the point (1,2) for the following stream function: $\psi = 0.6 (y^2-3x^2) + xy - 6$.
(c) What is orificemeter? Derive an equation to calculate the discharge using orificemeter.
(d) What is laminar sub-layer. Calculate displacement thickness for velocity distribution in boundary layer given by: $\frac{u}{U} = (\frac{y}{\delta})$?
(e) Define various dimensionless number and write its significance.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) Draw the Rheological graph of various fluids. Also explain rheological behavior of fluids.
(b) What are manometers? Explain vertical single column manometer.

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Derive the general form of continuity equation for 3-D flow. Also write its assumptions.
(b) Following data are given:- Barometer reading =740 mm of mercury, specific gravity of mercury =13.6, intensity of pressure =40 kPa. Calculate the intensity of pressure in S.I. units, both gauge and absolute.

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5. Attempt any *one* part of the following: 7 x 1 = 7

(a) A 30 cm diameter horizontal pipe terminates in a nozzle with the exit diameter of 7.51 cm if the water flows through the pipe at a rate of 0.15 m^3/sec. Calculate the force that will be exerted by the fluid on the nozzle?

(b) Explain loss of energy in pipe. Enlist various types of minor losses.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) The velocity distribution in boundary layer is given by: (U/V) = (3∈/2 - $\in^2$/2).Calculate the boundary layer thickness, momentum thickness & energy thickness. Where, $\in = \frac{y}{\delta}$

(b) What is boundary layer separation. Draw a neat diagram to explain.

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) The resulting force F of a supersonic plane during flight can be considered as dependent upon the length of aircraft"L", velocity"V", air viscosity "μ", air density"ρ" and the bulk modulus of air"K".Express the functional relationship between these variables and the resisting force.

(b) A pipe of diameter 1.5 m is required to transport an oil of specific gravity 0.9,viscosity $3x10^{-2}$poise, at the rate of 3000 liters/s. Test were conducted on a 15 cm diameter pipe using water at 20° C.Find the velocity and rate of flow in model. Viscosity of water at20° C = 0.01 poise.

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